

Point-in-Time Multi-Agent LLM Trading: A Reproducible LangGraph Stack with Regime-Conditioned Evaluation and Fair Backtesting

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Abstract

Large language models are increasingly used in financial decision pipelines, yet published agent stacks are often hard to reproduce and rarely evaluated under distinct market conditions on historical as-of dates. We present **Hindsight 20/20**, an open **LangGraph** stack with structured **Pydantic** outputs, point-in-time data access, optional **ticker anonymization**, and a fee-aware **paper backtest**. We evaluate the **full** pipeline on **RELIANCE.NS** and **TCS.NS** in **exogenous bear and bull windows** (per-ticker underlying return at most -10% or at least +10%), each from \$100,000 paper cash. In **both bear windows**, the agent loses less than buy-and-hold; in **bull windows**, outcomes diverge (RELIANCE lags B&H; TCS nearly tracks it). Frozen CSV artifacts reproduce tables and figures.

Keywords: algorithmic trading; large language models; multi-agent systems; reproducibility; point-in-time data; market regimes.

0.1 Introduction

Researchers are experimenting with LLMs inside trading workflows, but rarely report how the **same stack** behaves across **different underlying trends** or **more than one issuer**. We present **Hindsight 20/20**, an open **LangGraph** implementation with structured outputs, optional anonymization, a simulation end date, and a paper backtest. We evaluate the **full** pipeline on **RELIANCE.NS** and **TCS.NS** in four independent runs (bear and bull per ticker). Section 2 surveys related work; Section 3 the system; Section 4 results; Section 5 concludes; Section 6 lists contributions.

0.2 Related Work

LLM trading agents motivate multi-role graphs (Xiao et al., 2024; Yu et al., 2023; Yang et al., 2023). We emphasize reproducible engineering: one codepath, schema-constrained outputs, temporal controls, CSV artifacts.

Point-in-time data and leakage require explicit simulation horizons and optional anonymization.

Regime-split evaluation on the same implementation is uncommon; we report four frozen runs with **per-ticker calendars**, not regime prediction.

Execution realism via a paper ledger with stated fees.

0.3 Method / System

TradingAgentsGraph composes analysts, bull/bear **debate roles** (internal agents, distinct from market regimes), trader, and risk. Section 4 uses **PAPER_ABLATION=full**. **backtest_mvp.py --dates-csv** produces wide schedule CSVs.

0.4 Experiments

0.4.1 Regime definition

For each ticker, calendar windows in **regime-timeranges** are labeled **bear** if underlying buy-and-hold return is at most **-10%**, **bull** if at least **+10%** over the processed window. Each run starts from **\$100,000** fresh cash. Windows differ by ticker.

0.4.2 Protocol and results

Table 1. Four frozen runs.

Ticker	Regime	Window	Days	B&H	Agent	End equity	B/H/S
RELIANCE	Bear	2024-08-01 .. 2025-01-03	111	- 17.4%	-7.5%	92,459	7/28/76
RELIANCE	Bull	2025-01-01 .. 2025-06-03	108	+15.1%	+5.6%	105,559	9/25/74
TCS	Bear	2024-12-02 .. 2025-04-03	88	- 20.4%	-4.8%	95,166	12/24/52
TCS	Bull	2024-06-03 .. 2024-09-03	67	+21.9%	+19.6%	119,607	13/14/40

Bear (2/2): agent loses less than buy-and-hold. **Bull:** RELIANCE lags B&H; TCS nearly tracks B&H. Final signals remain SELL-heavy in most runs.

0.4.3 Figures

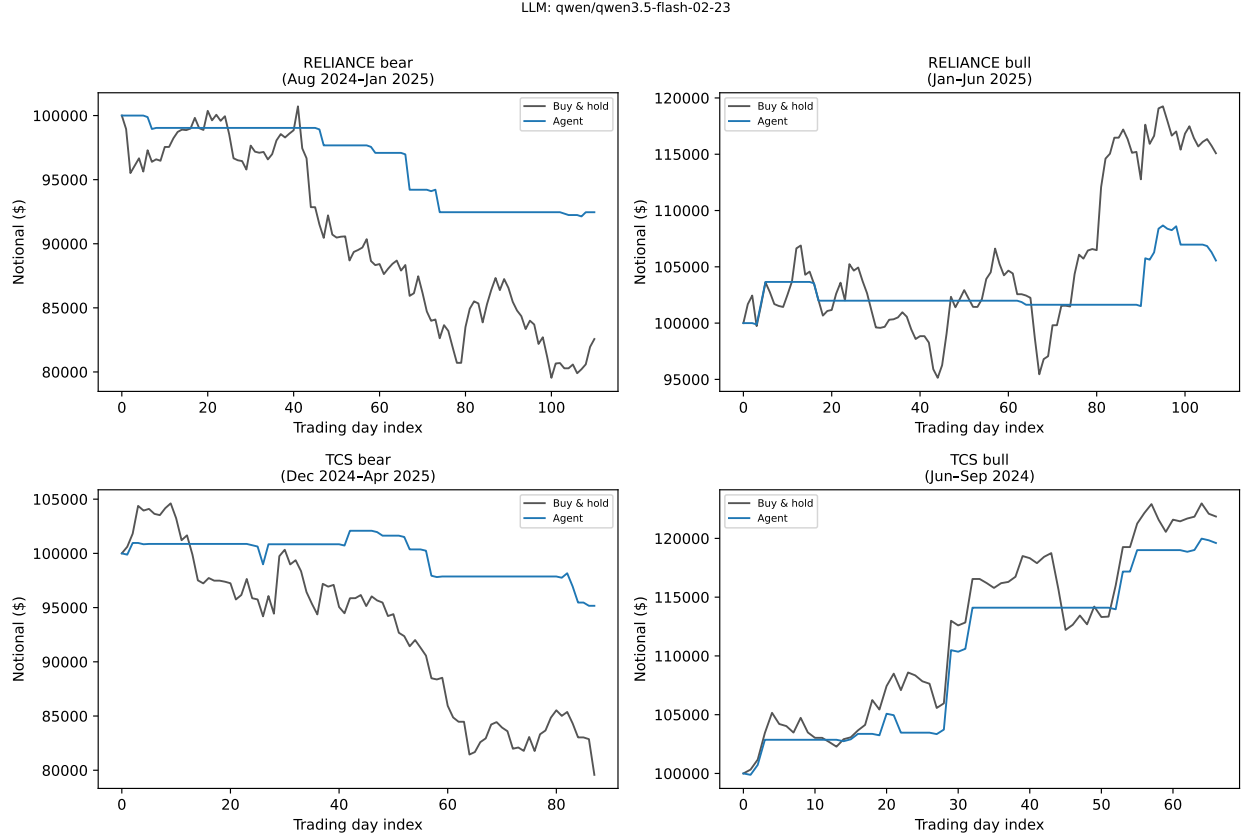


Figure 1: Figure 1. Agent vs buy-and-hold equity (2x2 grid).

0.5 Conclusion

We presented an open LangGraph stack and **multi-ticker regime-conditioned** evaluation. Bear windows show relative downside protection; bull outcomes are heterogeneous.

Limitations: two tickers, per-ticker calendars, one LLM.

0.6 Contributions

1. Multi-agent decision graph.

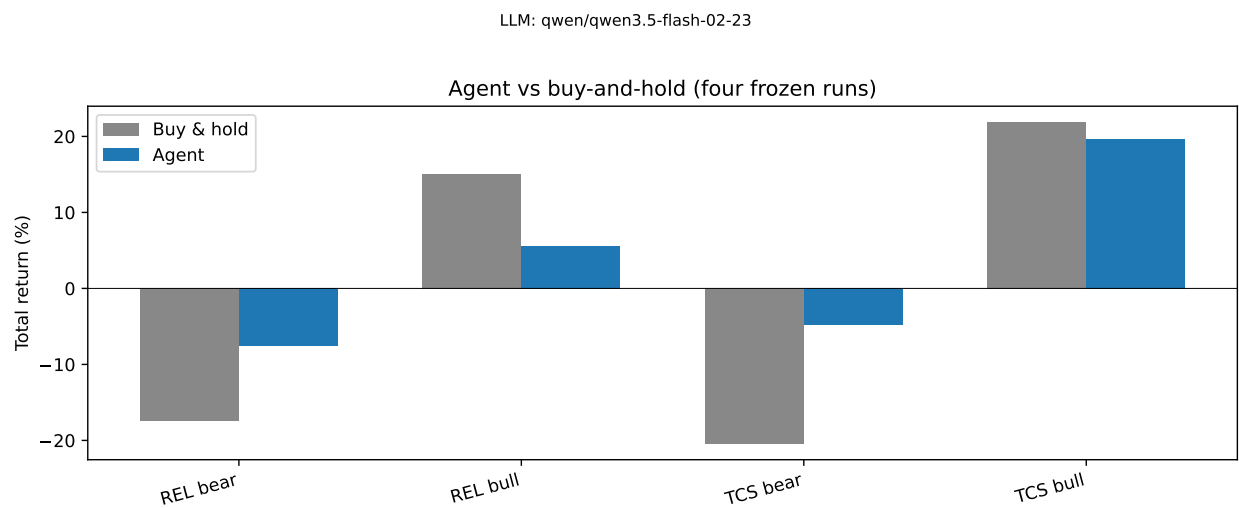


Figure 2: Figure 2. Total returns (four runs).

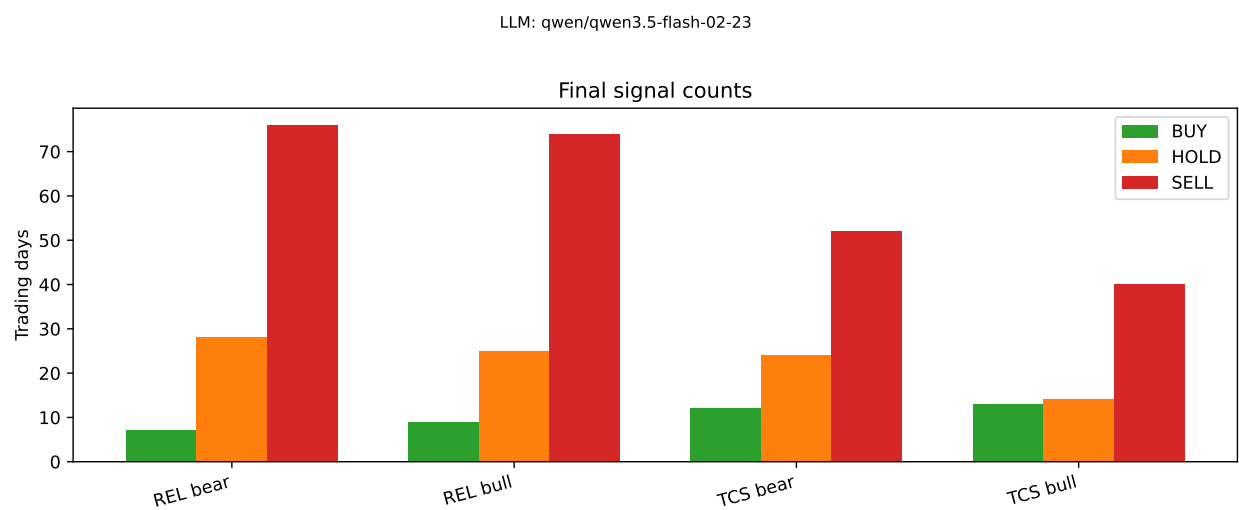


Figure 3: Figure 3. Signal counts.

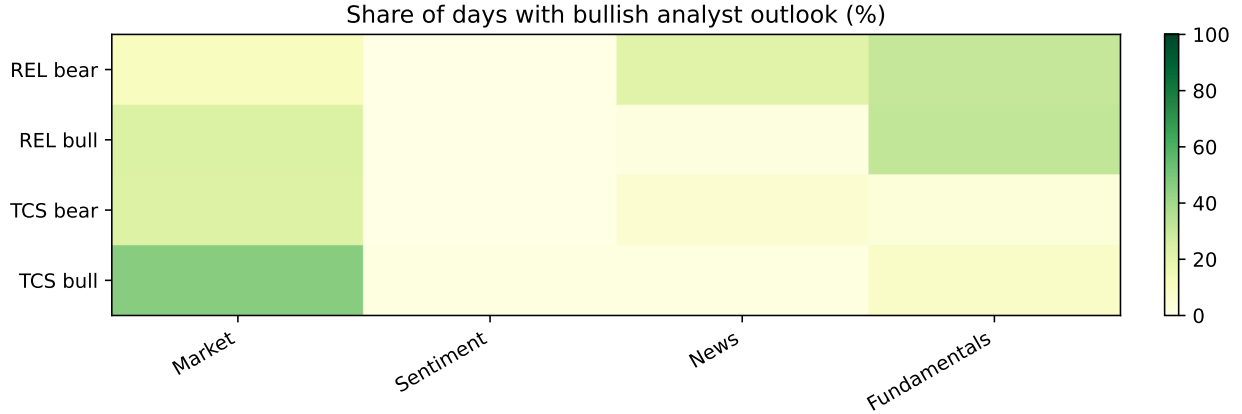


Figure 4: Figure 4. Bullish analyst-outlook share.

2. Multi-ticker regime-conditioned evaluation (four frozen CSVs).
3. Structured stage outputs.
4. Point-in-time policy and anonymization.
5. Backtest harness and analysis artifacts.

Acknowledgements

Not applicable.

References

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